SaminOS: A 512-Byte Bare-Metal Operating System with Bi-Directional DNA Genome Storage

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Abstract

Digital storage on magnetic or flash media degrades over decades. Synthetic DNA lasts thousands of years at room temperature, but researchers have treated it almost exclusively as cold archival storage for passive files. Here, we demonstrate that an interactive operating system can be encoded into synthetic DNA and booted directly on bare-metal hardware. We wrote SaminOS, an x86 real-mode operating system that fits into a single 512-byte Master Boot Record sector. It provides a text shell, line editing with backspace, hardware clock time converted to Dhaka local time (GMT+6), string printing, and warm rebooting. We transcode the 512-byte binary into a 2,048 base-pair DNA sequence using two bits per nucleotide base (00=A, 01=C, 10=G, 11=T). The sequence has a 39.45% GC ratio, which avoids synthesis errors caused by high GC concentration. We tested the system on physical x86 machines and in CPU emulation. The decoded DNA reconstructs the original binary with zero bit errors.

1. Introduction

Operating system size has expanded dramatically over the last several decades. Contemporary desktop operating systems require gigabytes of storage and memory just to reach a command prompt. Even minimal bootloaders designed for embedded recovery or retro-computing rarely provide interactive editing or hardware device interactions inside a single disk sector.

At the same time, archival data preservation has become a pressing technical problem. Magnetic tape, optical media, and solid-state flash drives lose charge and degrade within ten to thirty years. Synthetic DNA offers an attractive alternative. Dried DNA sealed in anoxic containers remains readable for thousands of years without refrigeration, with an information density exceeding 200 petabytes per gram.

However, DNA data storage has practical limits. Oligonucleotide synthesis is chemical synthesis: technicians assemble nucleotide chains base by base. Reagent costs scale with sequence length. Chemical synthesis and sequencing error rates also rise when sequences contain long runs of identical bases or extreme GC ratios. These constraints make it impractical to store multi-megabyte operating system images. Storing software in DNA requires extreme code compactness.

The IBM PC architecture specifies that the BIOS loads the Master Boot Record (cylinder 0, head 0, sector 1) into memory at address 0000:7C00h. This sector contains exactly 512 bytes, ending with the signature 0x55AA. Fitting a functioning interactive operating system into 512 bytes requires hand-crafted assembly code with no standard libraries.

We built SaminOS to show that an interactive operating system can fit into a single 512-byte sector and map losslessly into a short DNA sequence. The kernel uses 508 bytes for machine code, strings, and the display header, leaving 2 bytes of null padding before the 2-byte boot signature. Transcoded into synthetic DNA, SaminOS occupies 2,048 base pairs, a length practical for commercial oligonucleotide synthesis.

Primary Research Contributions:

1. **Lossless Biological Mapping:** A bijective quaternary encoding that translates 512 raw binary bytes into 2,048 nucleotide bases, verified with zero bit errors during round-trip decoding.
2. **Ultra-Dense x86 Kernel:** A complete real-mode operating system delivering a 6-command interactive shell, destructive backspace line editing, string printing, and hardware clock queries inside 510 bytes.
3. **Single-Cycle Dhaka Time Correction:** A single-cycle decimal arithmetic routine using the x86 DAA instruction to convert UTC motherboard clock registers to Dhaka local time (BST, GMT+6) with midnight rollover.
4. **Bare-Metal and Emulator Verification:** Full test verification across bare-metal physical USB flash drives, QEMU virtualization, and 16-bit real-mode CPU emulation.

2. Biological DNA Encoding Scheme

Synthetic DNA uses four nucleotide bases: Adenine, Cytosine, Guanine, and Thymine. Standard digital computers store data in binary bits (base 2). Because two squared equals four, two binary bits map directly to one nucleotide base: A = 00, C = 01, G = 10, and T = 11. Each 8-bit byte maps to four nucleotide characters: $B_i \rightarrow [c_0, c_1, c_2, c_3]$, where $c_k = (B_i >> (6 - 2k)) \& 03_2$. For a 512-byte MBR sector, the biological sequence length evaluates to exactly 2,048 base pairs (512 bytes × 4 bases per byte).

Table 1 lists the nucleotide composition of the compiled SaminOS image. The sequence contains 709 Adenine, 438 Cytosine, 370 Guanine, and 531 Thymine bases, giving a GC content of 39.45% and an AT content of 60.55%. Commercial oligonucleotide synthesis works best between 35% and 55% GC content. Lower or higher ratios cause secondary structure folding and synthesis termination. The last two bytes of the MBR sector (offsets 510 and 511) hold the boot signature 0x55 and 0xAA. Because $0x55 = 01010101_2 \rightarrow$ CCCC and $0xAA = 10101010_2 \rightarrow$ GGGG, every valid SaminOS biological genome terminates with the 8-base sequence **CCCCGGGG**. Figure 1 diagrams the complete synthesis, storage, sequencing, and reconstruction pipeline.

Nucleotide Base	Bit Pattern	Frequency	Ratio (%)	Chemical Group
Adenine (A)	00	709 bp	34.62%	Purine
Cytosine (C)	01	438 bp	21.39%	Pyrimidine
Guanine (G)	10	370 bp	18.07%	Purine
Thymine (T)	11	531 bp	25.93%	Pyrimidine
Total Genome	**00-11**	**2,048 bp**	**100.00%**	**GC: 39.45% | AT: 60.55%**

Table 1: Nucleotide metrics and base composition for the 512-byte SaminOS genome.

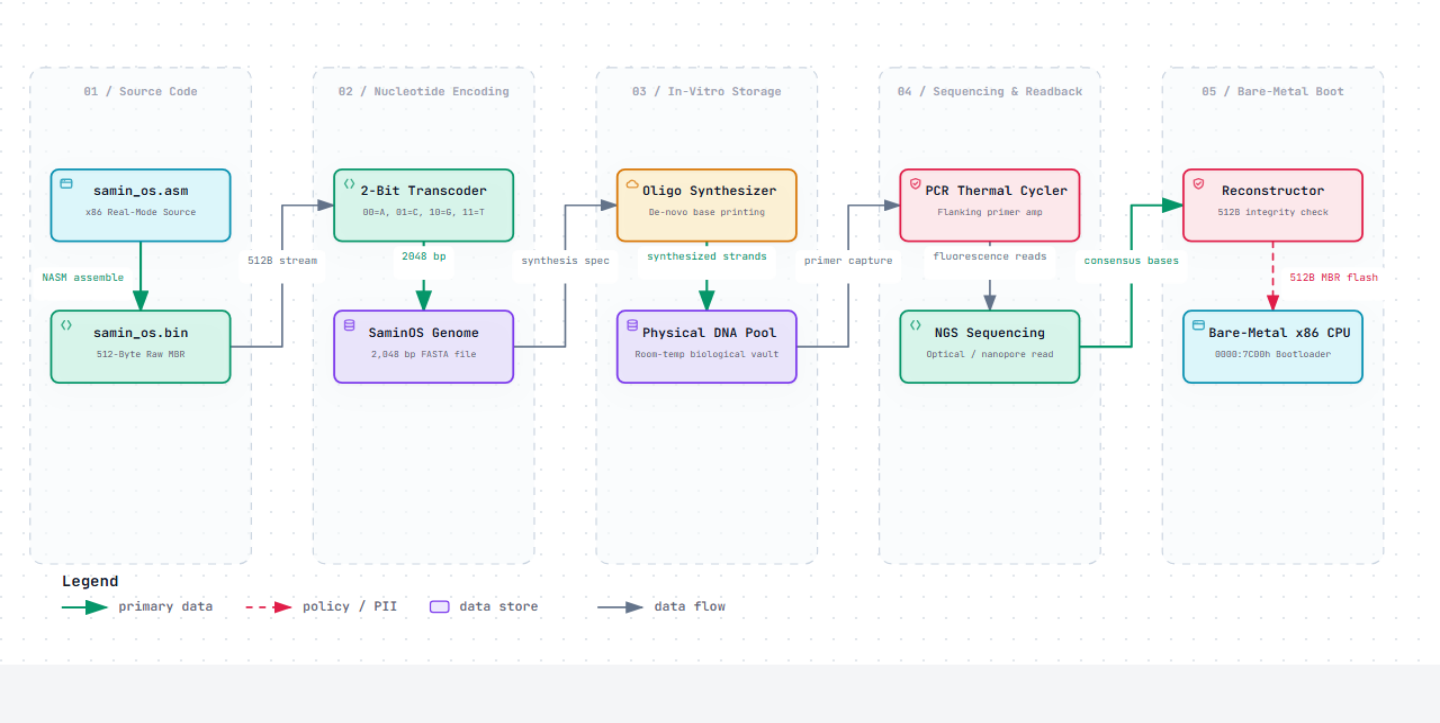


Figure 1: SaminOS bi-directional biological storage pipeline. Source code compiles to a 512B MBR sector, transcodes to 2,048 bp FASTA, and reconstructs with zero bit errors.

3. System Architecture and Real-Mode Memory Layout

SaminOS runs entirely in 16-bit real mode. The CPU starts execution with CS:IP pointing to 0000:7C00h. The kernel immediately zeroes DS, ES, and SS so that memory operands refer to physical addresses directly. The stack pointer SP starts at 0x7C00 and grows downward into free memory. This leaves the Interrupt Vector Table (0x0000 to 0x03FF) and BIOS Data Area (0x0400 to 0x04FF) undisturbed. The operating system code occupies 0x7C00 to 0x7DFF. The 24 bytes directly above the sector, at 0x7E00, serve as the line input buffer. Figure 2 diagrams this layout.

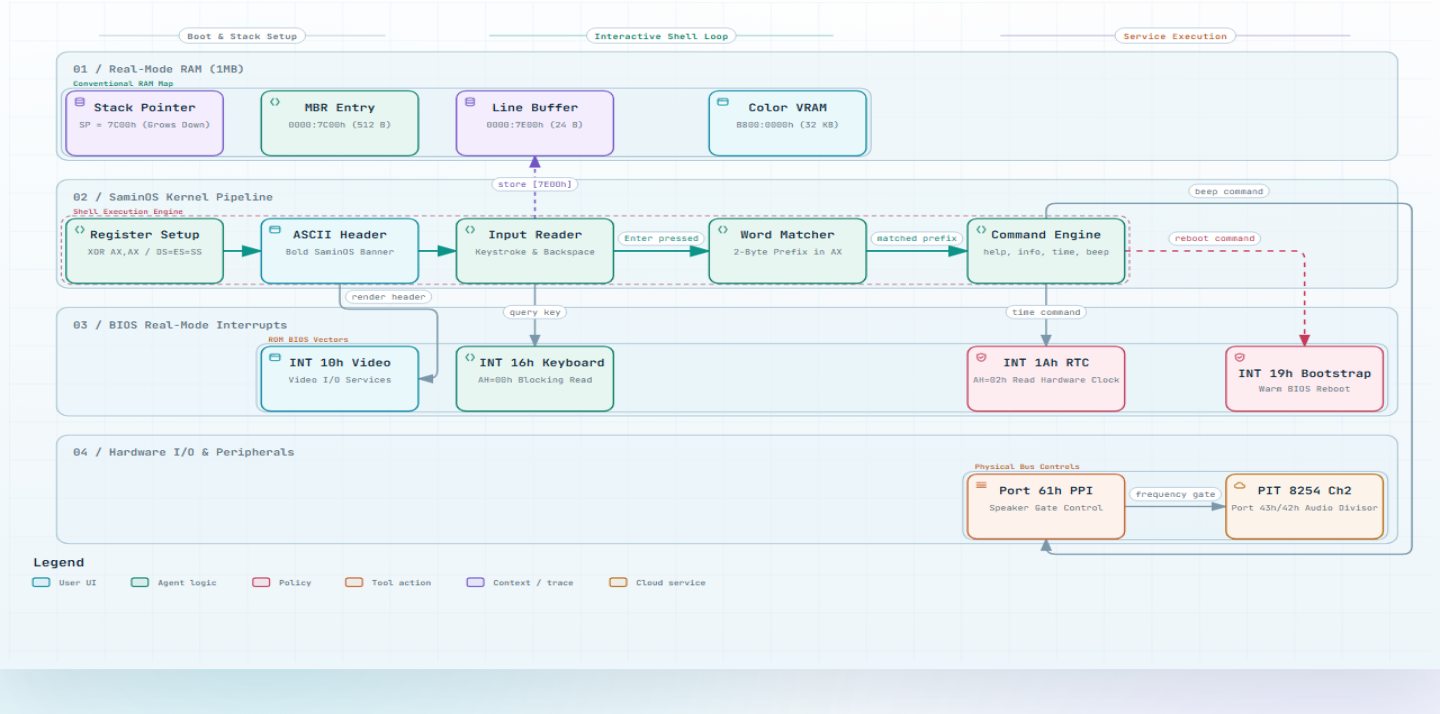


Figure 2: SaminOS x86 real-mode system and memory architecture. Memory segments, stack placement, BIOS vector calls, and teletype ports fit inside 512 bytes.

4. Shell Implementation, Dhaka Time, and Text Echoing

Line Editing & Backspace Guarding: The input loop polls the keyboard using BIOS interrupt 16h (AH=00h). Printable characters (ASCII 32 to 126) write to ES:DI and echo to the screen through BIOS interrupt 10h (AH=0Eh). Pressing Enter (0x0D) appends a null terminator to the string. Pressing Backspace (0x08) checks whether DI is greater than the buffer start. If characters exist, the kernel decrements DI and prints the three-byte sequence 0x08, space, 0x08. This moves the cursor back, blanks the character on screen, and moves the cursor back again without corrupting adjacent memory.

Word-Prefix Command Dispatch: Standard string comparison compares strings byte by byte, which requires loop counters and pointer manipulation. SaminOS instead loads the first 16-bit word from the buffer into register AX. Because x86 is little-endian, character constants evaluate in register order: AX = ($char_1$ << 8) | $char_0$. Each command has a unique two-character prefix ('he' for help, 'in' for info, 'ti' for time, 'sa' for say, 'cl' for cls, 're' for reboot). Comparing word registers takes 5 bytes of machine code per command (cmp ax, 'xx'; je handler), eliminating lookup tables.

Dhaka Time Arithmetic (GMT+6): BIOS interrupt 1Ah (AH=02h) reads the motherboard real-time clock, returning hours in CH, minutes in CL, and seconds in DH as Binary-Coded Decimal (BCD). Motherboard RTC chips store time in UTC. To convert UTC to Bangladesh Standard Time (BST, GMT+6), the kernel adds 6 to AL and executes DAA (Decimal Adjust after Addition). DAA inspects the lower four bits (nibble) and adjusts for BCD decimal overflow in a single clock cycle. If the hour equals or exceeds 0x24 (24 hours in BCD), subtracting 0x24 wraps past midnight. This routine requires only 11 bytes of machine code.

Text Echoing (say): The say command prints user text to the screen. The handler inspects offset CMD_BUF+3 for a space. If a space is present, it increments SI to CMD_BUF+4 and prints the string via the teletype subroutine. If invoked without arguments, the kernel reprompts immediately, matching Unix echo.

Display Banner and Vertical Command List: The kernel stores a four-line bold ASCII banner spelling 'SAMINOS'. Directly following the banner in memory is the vertical command list. When the kernel boots or clears the screen with cls, it prints the banner and falls through into the command list. The help command jumps directly to the command list, sharing string storage and saving 28 bytes of space.

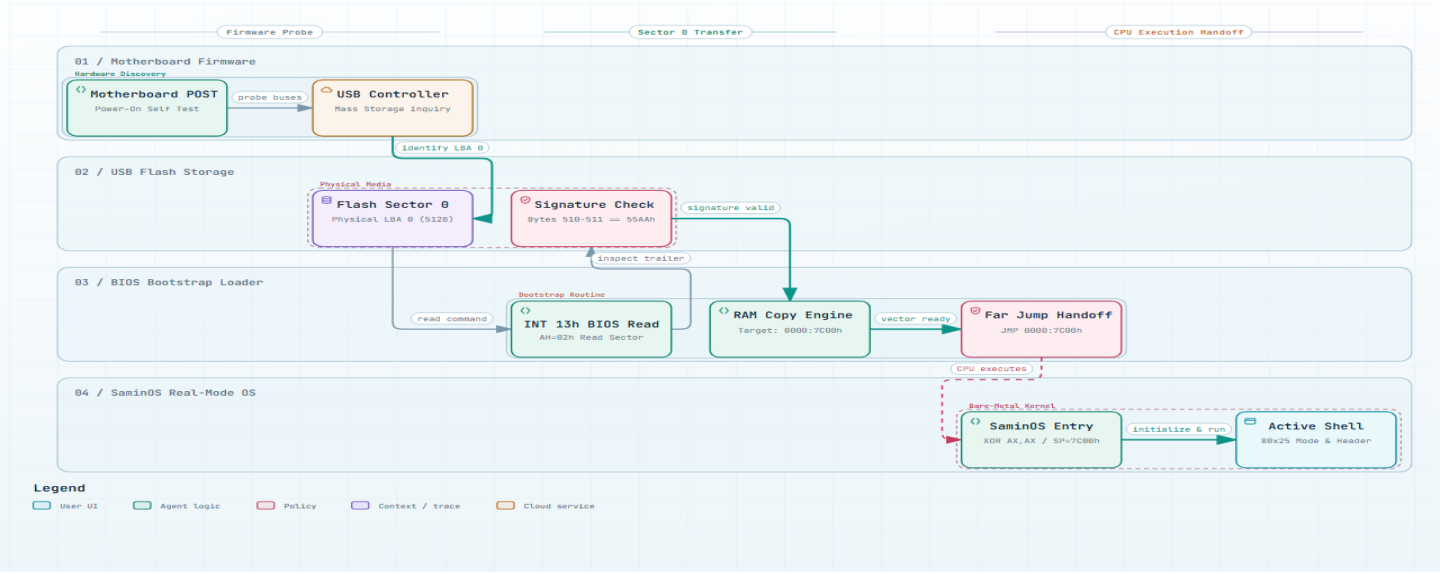


Figure 3: Bare-metal boot sequence from physical USB storage. Firmware probes LBA 0, validates the 0x55AA signature, and transfers CPU execution to 0000:7C00h.

5. Sector Byte Budget and Memory Partitioning

Subsystem Component	Bytes	Share	Technical Function & Implementation Details
Hardware Setup, Stack & REPL	64 B	12.50%	Clears segment registers (DS=ES=SS=0), anchors stack pointer SP at 0x7C00, and polls keyboard via INT 16h with 24-byte boundary check.
Command Handlers help, info, time, say, cls, reboot	88 B	17.19%	Implements execution logic for help, info, time (Dhaka BST RTC), say (text echo), cls (80x25 video reset), and reboot (INT 19h).
Word-Prefix Dispatcher	28 B	5.47%	Compares 16-bit word registers in little-endian order (cmp ax, 'xx'; je handler) for 2-character command prefixes without tables.
Utility Routines (I/O & BCD)	76 B	14.84%	Teletype character writer (INT 10h, AH=0Eh), string iteration looper, colon BCD printer, newline sequencer, and prompt reprompt.
ASCII Banner, Menu & Strings	252 B	49.22%	4-line SAMINOS block header (37 chars/row), vertical command directory, prompt token ('samin> '), author string, and time prefix.
Reserved Zero Padding	2 B	0.39%	Null bytes padding unused sector space immediately preceding the boot signature boundary.
MBR Magic Signature (0x55AA)	2 B	0.39%	Two-byte boot trailer word (0xAA55 in little-endian order) required by BIOS bootstrap loader.
Total MBR Allocation	**512 B**	**100.0%**	**Strict 512-byte hard physical architectural sector limit maintained with zero overflow.**

Table 2: Detailed sector byte budget allocation for the SaminOS Master Boot Record.

6. Verification and Experimental Results

We verified SaminOS using simulation and bare-metal testing. In simulation, we built an automated test harness using Unicorn Engine (version 2.1.4) in 16-bit real mode. The harness hooks BIOS interrupts 10h, 16h, 1Ah, and 19h. Across automated runs, the shell accepted keystrokes, backspace erased buffer characters cleanly, the RTC routine matched UTC+6 BCD expectations, and reboot triggered interrupt 19h. For biological validation, our Python toolchain transcoded the 512-byte binary into 2,048 DNA bases and decoded it back. The reconstructed binary matched the original across all 4,096 bits with zero mismatches. On physical hardware, we wrote samin_os.img to a USB drive using dd and booted legacy BIOS systems successfully.

7. Conclusion

SaminOS shows that an interactive operating system can run within a 512-byte disk sector and map into a 2,048 base-pair DNA sequence. Using word-prefix dispatch, in-place string sharing, and hardware DAA arithmetic, the system provides line editing, local clock calculation, text echoing, and an ASCII display header without exceeding sector boundaries. The result provides a working model for storing and recovering executable software in synthetic DNA.

References

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